

Academic and employer reputations agree mostly at the university level

Sora Yng

Abstract

Do the academic and the graduate labour markets rank university $\times$ field programmes alike, and at which level do they agree? We measure a programme’s academic reputation by its department’s rank in faculty hiring and compare it with its graduates’ pay in public data on 52 US fields and 33 UK subjects. Within a university, a department one standard deviation higher in its field’s hiring rank goes with about 0.8% higher pay (up to about 3% corrected for error in the rank); the 95% interval reaches zero, and the university-wide rank predicts pay at least as well. Across universities the two rankings agree (mean rank correlation +0.42), mostly in what both share with intake selectivity. In the UK, most agreement survives a control for graduates’ prior attainment, but what survives tracks intake selectivity at least as closely as the hiring rank. Agreement rises with years since graduation and is weak in UK subjects paid on national scales. Where the two markets’ rankings agree, they agree mostly at the university level.

1 Introduction

A student who applies to a university and a field chooses a programme: economics at one university, nursing at another. Much of what that choice promises is reputation. Admission cutoffs estimate what entry is worth: [Hoekstra \(2009\)](#) and [Anelli \(2020\)](#) compare applicants just above and just below the cutoff of one university, and [Kirkebøen et al. \(2016\)](#) use the cutoffs that ration entry to fields. Before such a return can be split into a university part and a department part, two descriptive facts must hold. Department standing must vary apart from university standing, and graduates’ pay must track that variation. We measure both.

The question matters because subject rankings compare departments, and rankings move applications ([Bowman and Bastedo, 2009](#); [Luca and Smith, 2013](#)). Strong research departments might teach different curricula, draw recruiters or be known to employers by name. If so, their graduates’ pay should track the department’s standing beyond the university’s. If not, a subject ranking that mostly restates the university’s rank may steer applicants between universities while appearing to compare programmes.

A university $\times$ field programme sells in two markets under one name, and its reputation in each is that market’s expectation of the quality of what it buys ([Tadelis, 1999](#); [MacLeod and Urquiola, 2015](#)). Hiring committees buy its doctoral graduates as faculty, and the net direction of their hires orders departments in a steep hierarchy ([Clauset et al., 2015](#)). We read that ordering, under the model’s assumptions, as *academic reputation*, measured by the field rank F and university-wide rank G of [Wapman et al. \(2022b\)](#). Employers buy its bachelor’s graduates, and what they pay reveals *employer reputation*, mixed with what they later learn about graduates and with who enrolls. Students’ admission choices reveal a third ranking, intake selectivity.

Among students admitted to similar colleges, the payoff to the more selective one is small on average ([Dale and Krueger, 2002, 2014](#)). The causal payoff to elite colleges concentrates in top jobs

and incomes (Zimmerman, 2019; Chetty et al., 2026), and the pay gradient in selectivity differs by field (Bloem et al., 2024). MacLeod et al. (2017) show that employers price a college’s reputation, measured by its graduates’ admission scores, and that the premium falls when individual skill becomes observable. We add a reputation revealed in a different market, faculty hiring, and ask whether it varies below the university and whether graduates’ pay follows it there.

We measure agreement by the rank correlation (*coupling*), across the universities that teach a field, between a department’s field rank and its graduates’ median earnings. Within a university, a department one standard deviation higher in F goes with 0.84% higher four-year pay [0.00, 1.64], and G predicts pay at least as well as F . Across universities the rankings agree: mean coupling is +0.419 [+0.355, +0.482] over 52 US fields, mostly in what both share with intake selectivity. Designs that compare more and less selective universities (Dale and Krueger, 2002; Anelli, 2020) largely vary university-level standing, on which, in US and UK public data, the two markets mostly agree. The analysis is descriptive. The model of Section 2 was written after most results were known; its tests, T0–T8, were specified before the analysis but after most other results had been seen, and not publicly registered (T1’s benchmark calibration was fixed after the literal one degenerated).

2 One name, two markets

Setup. Assumptions and proofs are in Appendix A. The unit is the programme (i, f) , over the institutions $\mathcal{I}_f$ with a bachelor’s programme and a ranked doctoral department in field f . Comparisons are across institutions within a field, so the model says nothing about the choice of field. In deviations from field means,

$$s_{if} = \alpha_f a_i + r_{if}, \quad \theta_{if} = \beta_f x_i + y_{if}, \quad x_i = \xi_i + v_i, \quad (1)$$

where s is the academy’s valuation of the department and θ the mean productivity of the programme’s graduates, selection included. The university-wide terms are a_i ($\text{Var } a = 1$) and intake ability ξ_i , as the admission criterion measures it; $v_i \perp \xi_i$ is the part of graduate quality that measured intake does not predict: unmeasured ability, value added, peer and resource effects. The field-specific terms r_{if} and y_{if} have correlation κ_f , and β_f prices x . We assume (A-orth) that r and y are uncorrelated with every university-wide variable.

Two reputations and a third ranking. *Academic reputation* is the position that the net direction of faculty hiring reveals, not a modelled belief about doctoral quality. In a gravity model of hires, the SpringRank score behind F (Wapman et al., 2022b; De Bacco et al., 2018) converges, to first order and up to scale, to $s^* = s + \pi/(2\beta)$, with π a department’s log ratio of doctorates produced to faculty hired (Proposition A1). Reading s^* as s requires volume neutrality (H0), which T0 audits. *Employer reputation* is the graduate market’s mean belief $V_{if}(t)$ about the productivity of the programme’s graduates t years after graduation. If employers compete and learn from graduates’ records (M1–M4; Lemma A3), log median pay is

$$m_{if}(t) = \mu_f(t) + V_{if}(t) + \beta_f g_i(t), \quad V_{if}(t) = (1 - \lambda_t) R_{if} + \lambda_t \theta_{if}, \quad (2)$$

with entry reputation $R = V(0)$, weight λ_t on graduates’ own records, and growth $g_i(t)$ with $g_i(0) = 0$ (Farber and Gibbons, 1996). Observed pay is thus employer reputation plus growth, net of the field mean. A university’s record pools all its graduates, so where employers see them often, $R \approx \theta$ at the university level (Lemma A2). Mean intake orders programmes as students do

if they all rank programmes alike and programmes admit by measured ability (S1; Lemma A4), which self-selection on gains (Kirkebøen et al., 2016) breaks. Both countries’ selectivity indices are university-level; our programme-level intake control is the UK attainment band.

Proposition 1 (What agreement measures) *Under A0, classical errors ($F = s^* + e_F$, $Y = m + e_Y$, $S = \xi + e_S$), A-orth and Lemmas A2 and A3, the Spearman coupling $\rho_f(t)$ increases with $\sqrt{\text{rel}_F \text{rel}_Y} \text{Corr}\{s^*, m(t)\}$, and*

$$\text{Cov}(s^*, m) = \underbrace{\frac{\text{Cov}(s^*, \xi) \text{Cov}(\xi, m)}{\text{Var } \xi}}_{\text{through measured intake}} + \underbrace{\alpha_f \text{Cov}(a^\perp \xi, m^U)}_{\text{university, beyond measured intake}} + \underbrace{\Lambda_{ft} \kappa_f \sigma_r \sigma_y}_{\text{field-specific}} + \underbrace{\frac{\text{Cov}(\pi^\perp \xi, m)}{2\beta}}_{\text{production}}, \quad (3)$$

with m^U the university-wide part of pay and $\Lambda_{ft} = \omega_f + (1 - \omega_f)\lambda_t$. The partial covariance given S is the last three terms plus $(1 - \text{rel}_S)$ times the first.

The first term is not selection alone: peer quality and resources that rise with intake also load on ξ (Dale and Krueger, 2002). A control for S removes the part of agreement that runs through measured intake, less a share $(1 - \text{rel}_S)$ left by error in S , whatever its cause, including students’ sorting on reputation (MacLeod and Urquiola, 2015).

Proposition 2 (University or department) *Let $c_X = \rho(X, Y)$ and $\cos \vartheta = \rho(F, G)$, and on standardised ranks let sr_F be the correlation of Y with the residual of F on G . Then $c_F = c_G \cos \vartheta + \text{sr}_F \sin \vartheta$, and F out-predicts G if and only if $\text{sr}_F > c_G \tan(\vartheta/2)$. If $G = a$ and H0 holds, sr_F prices only the field-specific term, attenuated by a reliability below rel_F .*

The field-specific term includes department standing and selection into the major (Bleemer and Mehta, 2022). A small sr_F means either that departments carry little that employers price or that employers do not attribute it to departments; these data cannot tell which (Appendix A.6).

3 Data and measurement

The academic ranking is the published SpringRank hiring rank of Wapman et al. (2022b), from a census of faculty at 368 PhD-granting US institutions in 2011–2020 (Wapman et al., 2022a). We use its field rank F_{if} and university-wide rank G_i , signed so that higher means higher standing. The pay ranking uses the median earnings Y of each programme’s bachelor’s graduates. US sources are College Scorecard medians four years after completion (3,444 programmes in 52 fields; U.S. Department of Education, 2026) and PSEO medians one, five and ten years after graduation for fixed cohorts (39 fields; U.S. Census Bureau, Longitudinal Employer-Household Dynamics, 2025, 2026). Scorecard medians cover only completers who received federal (Title IV) aid and who work and are not enrolled. PSEO (unemployment-insurance wage records) and UK LEO (tax records) lack the aid restriction; PSEO orders fields’ coupling much as Scorecard does (Spearman +0.68, 57 fields), but the within-institution slope rests on Scorecard alone. LEO medians are also released by graduates’ prior-attainment band (33 subjects, 134 providers; Department for Education, 2026). The UK has no field rank; its G is a SpringRank of 11,987 PhD-to-faculty moves in ORCID-derived records (Li, 2026). Intake selectivity is mean SAT and admission rate in the US, and the share of graduates with 360 or more UCAS points in the UK.

Partial coupling correlates the rank residuals of F and Y on the ranks of the controls. The reliability of F lies between 0.81 and 0.96. Brackets are 95% intervals that resample or cluster by

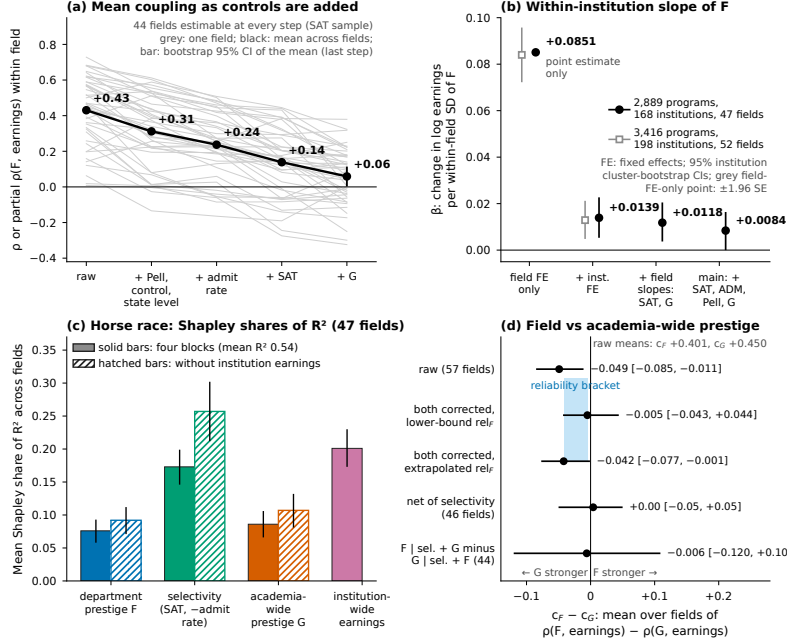


Figure 1: US coupling after university-level controls (Scorecard four-year pay; F and G are labelled “department” and “academia-wide prestige”). (a) Mean coupling as controls are added (44 fields; grey: single fields). (b) Within-institution slope of log pay per SD of F , from field fixed effects to the main specification. (c) Mean Shapley shares of R^2 , with and without (hatched) institution-wide earnings. (d) $c_F - c_G$: raw, corrected for error in F , and net of selectivity. 95% institution-bootstrap intervals.

institution (PSEO: two-way field and institution clusters; UK: subjects, then providers).¹ SI Sn refers to the supplementary information of the full-length paper. Two tests qualify what F measures. T0 finds that F partly reflects doctoral production: residualised on production, F keeps 0.598 [0.471, 0.725] of its coupling, a lower bound because the residualisation also removes net export that valuation itself produces. At most about two fifths of coupling thus runs through production; the within-institution slope has not been re-estimated net of it. T1 finds that F does not order departments alike as placers and as hirers of PhDs, so we read it as a net-placement position, not one index of doctoral quality.

4 Results

4.1 The two rankings agree, to a degree that varies by field

Across the 52 US fields, mean coupling of F with four-year pay is +0.419 [+0.355, +0.482] (SI S3). It lies more than 1.96 bootstrap standard errors above zero in 39 fields and below zero in none, from +0.71 in computer science to +0.07 in nursing. The field map follows how closely each field’s pay tracks mean SAT (true-score correlation +0.94, 23 fields), as it would if fields differed only in the scale at which they price one university-level index (Corollary A1; a consistency check, not

¹The canonical sample is the 52 fields with at least 15 matched institutions; heterogeneity, reliability and F -against- G statistics use the 57 with at least 8. Other samples: the control ladder, 44 fields estimable at every step; the within-institution slope and Shapley shares, 47 with SAT data; the career analysis, 39 in the balanced PSEO panel; the field-map check, the 23 largest. UK: 33 subjects with at least 15 providers, 28 for same-band comparisons.

a test). It therefore cannot separate a department gradient in pay from a gradient in university status.

4.2 Agreement sits at the university level and is largely shared with intake selectivity

Mean SAT and admission rate alone lower mean US coupling from +0.43 to +0.14 (47 fields); Pell share, control and state earnings level alone lower it to +0.31 (Fig. 1a). In per-field rank regressions, selectivity's mean Shapley share of R^2 is 0.173, against 0.076 for F , 0.086 for G and 0.201 for institution-wide earnings, which contain the field's own graduates (Fig. 1c).

Before any control, G couples with pay slightly more strongly than F does: $c_F - c_G = -0.049$ $[-0.085, -0.011]$ (57 fields; Fig. 1d). At the across-field means, sr_F is 0.077 against a threshold of 0.156 (field by field they average 0.055 and 0.141), and F out-predicts G in only 15 of 57 fields (Proposition 2). Corrected for error in F , the difference is -0.005 $[-0.043, +0.044]$ at the lower-bound reliability; under either correction, and net of selectivity, F does not detectably out-predict G , though a small advantage is not excluded.

Within universities, the department's part of the agreement is small. With field fixed effects alone, one within-field standard deviation (SD) of F goes with +0.084 log points of four-year pay; institution fixed effects lower this to +0.013 (Fig. 1b). The main specification adds field-specific slopes on mean SAT, admission rate, Pell share and G ; one SD of F then goes with +0.0084 log points $[+0.0000, +0.0164]$, about 0.84% (2,889 programmes; $[-0.0012, +0.0180]$ under the two-way variance; about zero with one- and five-year medians, which describe other cohorts). The upper 95% limit is 1.6–1.8% per SD of the observed rank. The controls absorb most of F , and the part they leave has a reliability of about 0.3 at the lower bound; corrected for it, the slope is about 3% (+0.0297), so the department's part is diluted, not shown to be absent. Because US selectivity is measured for the university, the slope also carries any sorting of stronger students into a university's higher-ranked departments, of unknown sign; a control for the programme's own Pell share does not lower it, and comparing majors that restrict entry with open majors is too weak to separate the two (SI S4). Only computer science has an individually clear remainder, +0.050 log points per SD $[+0.012, +0.096]$, pooled over horizons (128 programmes; the sample lacks 21 of the 50 highest-ranked departments, among them the University of California campuses).

4.3 In the UK, most agreement survives a control for graduates' prior attainment

The UK data add pay by graduates' own prior attainment, but no field rank, so they bear on the university level only. Coupling of G with five-year pay averages +0.486 $[+0.398, +0.559]$ across 33 subjects. Among graduates of the same attainment band coupling is +0.227, against +0.342 for all graduates of the same providers, so 66% [51, 79] survives (28 subjects; 87% [74, 100] on 27 subjects after correcting for noise in band medians). The band comparison controls graduates' own attainment, whereas institution-mean selectivity also removes peer and resource effects: net of institution-level selectivity instead, UK coupling keeps 26% (+0.131), a fall comparable to the US one. If the UK G is measured with little error, what remains within bands tracks intake selectivity at least as closely as G . Selectivity keeps a same-band partial coupling of +0.252 $[+0.172, +0.310]$ net of G , while G net of selectivity, itself a status measure, is +0.033, not distinguishable from zero (28 subjects). At a validity of 0.74, however, the data allow G 's same-band partial to reach +0.26.

4.4 Agreement rises with years since graduation and is weak under national pay scales

Within fixed US graduation cohorts, coupling rises by $+0.031$ [$+0.021$, $+0.041$] per year since graduation (39 fields; PSEO), and UK coupling rises by about as much. The rise mixes career time with calendar time, and about three fifths of the US rise is shared with entry-year selectivity. Graduate study is not excluded: if graduates of higher-status universities more often leave the earnings data to study and return with higher pay, coupling rises mechanically. With coverage and the university’s research-doctorate rate as controls the slope falls from $+0.0305$ to $+0.0103$ per year, though that rate tracks G ; coverage alone removes about a fifth, and a fifth to a third of the rise moves with where graduates work (T7). Within the model, a rise at well-known universities needs earnings growth, falling pay variance unrelated to F , or learning about departments, which is small since sr_F is (Proposition A2); outside it, graduate study, coverage and migration can produce the same rise.

Coupling is weak in UK subjects paid largely on national scales: nursing, medicine and teaching couple at $+0.120$, against $+0.523$ for the other subjects (difference -0.404 [-0.566 , -0.238]). Allied health, much of it also paid on national bands, is an exception at $+0.570$, as is US special education ($+0.55$). A schedule that only rescales market pay leaves correlations unchanged (Proposition A2), so weak coupling reflects noisier medians, schedule variance unrelated to status or weaker market agreement; these data do not separate them.

5 Discussion

For a student choosing among university $\times$ field programmes, a department’s academic reputation predicts little of graduates’ pay beyond the university’s rank and intake: about 0.8% per SD, with an upper 95% limit of 1.6–1.8%, or about 3% once corrected for error in the rank. Computer science may be the exception. Subject rankings built on academic standing may share this limit: F plays that part here, and Britton et al. (2022) find a similar pattern for UK league tables, none of whose subject ratings correlates positively and significantly with returns net of intake.

Agreement is not a return: the university terms in (3) can be pure common causes. At an admission cutoff, the model splits the effect of admission to programme p over a fallback p' in the same field into

$$\tau(t) = \Delta\delta + (1 - \lambda_t)(\Delta R - \Delta\delta) + \beta_f \Delta g(t) \quad (4)$$

(Corollary A3): value added, a reputation rent that employers pay even for a pure signal and that learning erodes (Altonji and Pierret, 2001; Arcidiacono et al., 2010; MacLeod et al., 2017), and growth. Cutoff data can answer two questions. First, in data like those of Anelli (2020), does the admission effect decline with years since graduation, a rent that employers learn away, or rise through growth? The profile of $\tau(t)$ separates the two (Corollary A3(iii)); rising cross-sectional coupling does not. Second, across many programme cutoffs, does the effect differ with the fallback’s department-rank gap, among fallbacks at universities of similar standing?

One university’s cutoff answers only the first: its rejected applicants go elsewhere, so it contrasts universities, and the changed field choice through which part of Anelli’s premium runs is a channel within that contrast. The second needs ranked application lists, which identify each fallback (Kirkebøen et al., 2016) and reveal students’ own ordering of programmes (Avery et al., 2013), and a department rank where cutoffs exist. F exists only for the US, which lacks programme cutoffs, but a hiring rank can be built from ORCID-derived moves, as our UK G was. Because F and G correlate 0.786 on average, the department-rank gap varies little given the university-rank gap, so

many cutoffs are needed. Projected on the target-minus-fallback gaps in the two reputations and in peer ability, cutoff estimates of τ would show how much of a return goes with the university’s standing and how much with the department’s; the model has no channel from academic standing to pay, so these loadings describe association across cutoffs, not causal channels (Appendix A.9). Earnings must span a decade or more: first-year medians are distorted by graduate study and coverage, and the rent should erode while growth may rise. Which reputation students themselves follow when they choose a programme is the next study’s question.

Limitations. The analysis uses public aggregates and does not separate value added, selection and employer screening, nor say why agreement below the university is weak. It assumes, without testing, that pay equals expected productivity. Programme medians miss the upper tail, where elite returns concentrate (Zimmerman, 2019; Chetty et al., 2026), and PSEO lacks most of the top quartile of ranked institutions. Pay is not netted of where graduates work: only raw five-year PSEO coupling was checked net of the wage level of graduates’ destination census divisions (unchanged, T7); the Scorecard ladder, partial couplings and within-institution slope were not, and location within divisions is never netted. Selectivity is measured for the university, not the programme, and whatever the proxies miss stays in every partial coupling. About a fifth of pay coupling runs between employer sectors, which reflects employers’ valuation only if employers ration access to sectors. The UK hierarchy rests on self-reported ORCID careers of uncertain validity. The model’s tests share data with results already seen, and two of them are inconclusive for lack of power.

References

- Atila Abdulkadiroğlu, Parag A. Pathak, Jonathan Schellenberg, and Christopher R. Walters. Do parents value school effectiveness? *American Economic Review*, 110(5):1502–1539, 2020. doi: 10.1257/aer.20172040.
- Joseph G. Altonji and Charles R. Pierret. Employer learning and statistical discrimination. *The Quarterly Journal of Economics*, 116(1):313–350, 2001. doi: 10.1162/003355301556329.
- Massimo Anelli. The returns to elite university education: A quasi-experimental analysis. *Journal of the European Economic Association*, 18(6):2824–2868, 2020. doi: 10.1093/jeea/jvz070.
- Peter Arcidiacono, Patrick Bayer, and Aurel Hizmo. Beyond signaling and human capital: Education and the revelation of ability. *American Economic Journal: Applied Economics*, 2(4):76–104, 2010. doi: 10.1257/app.2.4.76.
- Christopher N. Avery, Mark E. Glickman, Caroline M. Hoxby, and Andrew Metrick. A revealed preference ranking of U.S. colleges and universities. *The Quarterly Journal of Economics*, 128(1):425–467, 2013. doi: 10.1093/qje/qjs043.
- Zachary Bleemer and Aashish Mehta. Will studying economics make you rich? A regression discontinuity analysis of the returns to college major. *American Economic Journal: Applied Economics*, 14(2):1–22, 2022. doi: 10.1257/app.20200447.
- Michael D. Bloem, Xiaowen Hu, and Michael Hurwitz. Understanding variation in post-college earnings: Evidence from the U.S. Department of Education’s College Scorecard. EdWorkingPaper 24-1112, Annenberg Institute at Brown University, 2024. URL <https://doi.org/10.26300/9ykq-pg87>.

- Nicholas A. Bowman and Michael N. Bastedo. Getting on the front page: Organizational reputation, status signals, and the impact of U.S. News and World Report on student decisions. *Research in Higher Education*, 50(5):415–436, 2009. doi: 10.1007/s11162-009-9129-8.
- Jack Britton, Laura van der Erve, Chris Belfield, Anna Vignoles, Matt Dickson, Yu Zhu, Ian Walker, Lorraine Dearden, Luke Sibiet, and Franz Buscha. How much does degree choice matter? *Labour Economics*, 79:102268, 2022. doi: 10.1016/j.labeco.2022.102268.
- Raj Chetty, David J. Deming, and John N. Friedman. Diversifying society’s leaders? The determinants and causal effects of admission to highly selective private colleges. *The Quarterly Journal of Economics*, 141(1):51–145, 2026. doi: 10.1093/qje/qjaf050. Earlier version: NBER Working Paper 31492 (2023).
- Aaron Clauset, Samuel Arbesman, and Daniel B. Larremore. Systematic inequality and hierarchy in faculty hiring networks. *Science Advances*, 1(1):e1400005, 2015. doi: 10.1126/sciadv.1400005.
- Stacy B. Dale and Alan B. Krueger. Estimating the payoff to attending a more selective college: An application of selection on observables and unobservables. *The Quarterly Journal of Economics*, 117(4):1491–1527, 2002. doi: 10.1162/003355302320935089.
- Stacy B. Dale and Alan B. Krueger. Estimating the effects of college characteristics over the career using administrative earnings data. *Journal of Human Resources*, 49(2):323–358, 2014. doi: 10.3368/jhr.49.2.323.
- Caterina De Bacco, Daniel B. Larremore, and Cristopher Moore. A physical model for efficient ranking in networks. *Science Advances*, 4(7):eaar8260, 2018. doi: 10.1126/sciadv.aar8260.
- Department for Education. Graduate outcomes (LEO): Provider level data, 2022-23. <https://explore-education-statistics.service.gov.uk>, 2026. Accessed 2026-06-22.
- Henry S. Farber and Robert Gibbons. Learning and wage dynamics. *The Quarterly Journal of Economics*, 111(4):1007–1047, 1996. doi: 10.2307/2946706.
- David Gale and Lloyd S. Shapley. College admissions and the stability of marriage. *The American Mathematical Monthly*, 69(1):9–15, 1962. doi: 10.1080/00029890.1962.11989827.
- Bruce C. Greenwald. Adverse selection in the labour market. *The Review of Economic Studies*, 53(3):325, 1986. doi: 10.2307/2297632.
- Justine S. Hastings, Christopher A. Neilson, and Seth D. Zimmerman. Are some degrees worth more than others? Evidence from college admission cutoffs in Chile. NBER Working Paper 19241, National Bureau of Economic Research, 2013. URL <https://doi.org/10.3386/w19241>.
- Mark Hoekstra. The effect of attending the flagship state university on earnings: A discontinuity-based approach. *The Review of Economics and Statistics*, 91(4):717–724, 2009. doi: 10.1162/rest.91.4.717.
- Lisa B. Kahn. Asymmetric information between employers. *American Economic Journal: Applied Economics*, 5(4):165–205, 2013. doi: 10.1257/app.5.4.165.
- Lars J. Kirkeboen, Edwin Leuven, and Magne Mogstad. Field of study, earnings, and self-selection. *The Quarterly Journal of Economics*, 131(3):1057–1111, 2016. doi: 10.1093/qje/qjw019. Erratum: *Quarterly Journal of Economics* 132(3), 1551–1552 (2017), doi:10.1093/qje/qjx025.

- William H. Kruskal. Ordinal measures of association. *Journal of the American Statistical Association*, 53(284):814–861, 1958. doi: 10.1080/01621459.1958.10501481.
- Fabian Lange. The speed of employer learning. *Journal of Labor Economics*, 25(1):1–35, 2007. doi: 10.1086/508730.
- Yifeng Li. ORCID-derived academic mobility edges with resolved cities and organizations. Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.19651302>. Version 20260419.
- Michael Luca and Jonathan Smith. Salience in quality disclosure: Evidence from the U.S. News college rankings. *Journal of Economics & Management Strategy*, 22(1):58–77, 2013. doi: 10.1111/jems.12003.
- W. Bentley MacLeod and Miguel Urquiola. Reputation and school competition. *American Economic Review*, 105(11):3471–3488, 2015. doi: 10.1257/aer.20130332.
- W. Bentley MacLeod, Evan Riehl, Juan E. Saavedra, and Miguel Urquiola. The big sort: College reputation and labor market outcomes. *American Economic Journal: Applied Economics*, 9(3): 223–261, 2017. doi: 10.1257/app.20160126. Earlier version: NBER Working Paper No. 21230, June 2015.
- Jack Mountjoy and Brent Hickman. The returns to college(s): Relative value-added and match effects in higher education. NBER Working Paper 29276, National Bureau of Economic Research, 2021. URL <https://doi.org/10.3386/w29276>.
- Uta Schönberg. Testing for asymmetric employer learning. *Journal of Labor Economics*, 25(4): 651–691, 2007. doi: 10.1086/522905.
- Steven Tadelis. What’s in a name? Reputation as a tradeable asset. *American Economic Review*, 89(3):548–563, 1999. doi: 10.1257/aer.89.3.548.
- U.S. Census Bureau, Longitudinal Employer-Household Dynamics. Post-Secondary Employment Outcomes (PSEO). https://lehd.ces.census.gov/data/pseo_experimental.html, 2025. Release V4.13.0 / 2025Q4; accessed 2026-06.
- U.S. Census Bureau, Longitudinal Employer-Household Dynamics. Post-Secondary Employment Outcomes (PSEO), release V4.14.1. https://lehd.ces.census.gov/data/pseo_experimental.html, 2026. Release V4.14.1 / 2026Q2; accessed 2026-09-23.
- U.S. Department of Education. College Scorecard data: Field of study, institution and historical (all data files) releases of 10 june 2026. <https://collegescorecard.ed.gov/data/>, 2026. Accessed 2026-06-20 (Field of Study), 2026-09-23 (Institution) and 2026-09-24 (historical raw-data archive).
- Michael Waldman. Job assignments, signalling, and efficiency. *The RAND Journal of Economics*, 15(2):255, 1984. doi: 10.2307/2555679.
- K. Hunter Wapman, Sam Zhang, Aaron Clauset, and Daniel B. Larremore. Data for “Quantifying hierarchy and dynamics in U.S. faculty hiring and retention”. Zenodo dataset, version 1, 2022a. URL <https://doi.org/10.5281/zenodo.6941651>. Published SpringRank ranks for 81 fields (107 fields in the census), 368 PhD-granting institutions, 2011–2020; CC BY 3.0 US.

K. Hunter Wapman, Sam Zhang, Aaron Clauset, and Daniel B. Larremore. Quantifying hierarchy and dynamics in US faculty hiring and retention. *Nature*, 610(7930):120–127, 2022b. doi: 10.1038/s41586-022-05222-x. Author Correction: *Nature* 619, E49 (2023), doi:10.1038/s41586-023-06379-9.

Seth D. Zimmerman. Elite colleges and upward mobility to top jobs and top incomes. *American Economic Review*, 109(1):1–47, 2019. doi: 10.1257/aer.20171019.

A Proofs for Section 2

Moments are taken within field f over $i \in \mathcal{I}_f$. $X^{\perp Z}$ is the residual of X on Z , and $\sigma_X^2 = \text{Var } X$. The university level is called the name. The tests T0–T8 were specified before the analysis; they are not publicly registered.

Table 1: Three revealed rankings of the same programmes. Audits: tests specified before the analysis; not publicly registered.

Evaluator	Revealing choice	Indicator	Identifying assumption	Audit
Hiring committees (academic market)	net direction of faculty hires	F_{if} field rank, G_i university-wide rank (Wapman et al., 2022b)	gravity model (H2); H0 volume neutrality reads s^* as the valuation s	T0 partly production; T1 not one index
Employers (graduate market)	wages	$Y_{if}(t)$, programme median log earnings t years after graduation	M1 pay equals the posterior mean of productivity; M2 institution and major seen; M3 no pay schedule; M4 median covers market workers	T7 location-robust; T8 between-sector share 0.21
Students	applications and matriculation under capacity	S_i , mean SAT and admission rate (UK: share of graduates with ≥ 360 UCAS points; LEO earnings by prior-attainment band within provider $\times$ subject)	S1 common ranking, admission by measured ability; defined for programmes only where admission is by programme (UK UCAS, Norway, Chile, China)	none: the least audited leg

A.1 Assumptions

A0 (Gaussian copula). The model holds for the normal scores of the latent components and the indicators, which are jointly Gaussian. For two such variables with Pearson correlation r , Spearman’s $\rho = \frac{6}{\pi} \arcsin(r/2)$ (Kruskal, 1958), which is strictly increasing. Statements about signs and orderings therefore hold for ρ ; magnitudes are statements about normal scores. Near zero, a rank-based partial correlation need not have the sign of the latent one.

E (classical errors). $F = s^* + e_F$ (on the scale of s^* , Proposition A1), $G = a + e_G$, $Y(t) = m(t) + e_Y$ and $S = \xi + e_S$, with errors of mean zero, independent of one another and of every latent variable; rel_X is the latent share of $\text{Var } X$. The production component that T0 finds is part of s^* , not of e_F . S and G aggregate an institution’s programmes, so e_S contains $w_{if}y_{if}$ and e_G a hire-weighted share of r_{if} , with w_{if} the programme’s share of its institution; E holds up to terms of that order.

A-orth (no covariance across levels). r_{if} and y_{if} are uncorrelated with every university-wide variable: a_i , ξ_i , v_i , $g_i(t)$ and employers’ name belief $\hat{x}_i$, up to terms of order $\max_g w_{ig}$ (Lemma A2). Defining r and y as within-field residuals delivers $\text{Cov}(a, r) = \text{Cov}(x, y) = 0$ and nothing more; the rest is assumed. It fails if departments are relatively strong at selective institutions ($\text{Cov}(r, \xi) > 0$) or if high-status institutions draw relatively able students into the major ($\text{Cov}(a, y) > 0$).

H0–H2 (academic market). A_{uv} counts field- f faculty at v with a doctorate from u ($u \neq v$).

Gravity: $\mathbb{E}A_{uv} = p_u h_v \varphi(s_u - s_v) \zeta_{uv}$, with origin and destination parameters $p_u, h_v > 0$, φ positive, strictly increasing and differentiable, $\zeta_{uv} = \zeta_{vu} \geq 0$ (geography, subfield overlap), and no ties in s . Let $\pi_u = \log(p_u/h_u)$. The parameters p, h are not the observed volumes: the expected number of u 's doctorates placed, $p_u \sum_v h_v \varphi(s_u - s_v) \zeta_{uv}$, also depends on s . H0 (volume neutrality): π_u is the same for all departments. H1 (vertical sorting): for every pair with $\zeta_{uv} > 0$, $s_u > s_v$ implies $\mathbb{E}A_{uv} > \mathbb{E}A_{vu}$. H2 (many hires): $\mathbb{E}A_{uv} = n\alpha_{uv}$ with α fixed and $A_{uv}/n \rightarrow_p \alpha_{uv}$ as $n \rightarrow \infty$. Under the gravity form H0 implies H1, since $\mathbb{E}A_{uv}/\mathbb{E}A_{vu} = \exp(\pi_u - \pi_v) \varphi(\Delta)/\varphi(-\Delta)$ with $\Delta = s_u - s_v$, and $\varphi(\Delta) > \varphi(-\Delta)$ for $\Delta > 0$.

M1–M4 (graduate market). Graduate k of programme $p = (i, f)$ has productivity $\eta_k = \theta_p + \varepsilon_k$, $\varepsilon_k \sim N(0, \sigma^2)$ independent, plus growth $\beta_f g_i(t)$. M1: employers compete, and pay is the posterior mean of productivity given their information (Farber and Gibbons, 1996): the credential, the graduate's own signals (precision h_b at hiring, h_ε in each later year), a public signal P_i about the name, and records of past graduates' performance attributed to units. Attribution is costly or markets are segmented, so a unit's record counts the graduates a local market has attributed to it. M1 includes four conditions. (i) Learning is symmetric: employers competing in a local market share the same records and signals; with employer-specific information the competitive wage need not be the posterior mean (Waldman, 1984; Greenwald, 1986; Schönberg, 2007; Kahn, 2013). (ii) The prior precision h_0 of η_k given the programme's record is the same across programmes. This is an approximation for when σ^2 dominates the prior variance of θ given the record, $\beta_f^2(1 - \omega_i)\tau_i^2 + (1 - \omega_f)\tau_y^2$, which varies with the weights of Lemma A2. (iii) Growth is known to employers and $g_i(0) = 0$. (iv) ω_{if} is constant within a field, or independent of r and y (programme size, which sets the attributed record, may vary with status). M2: the institution and the major are seen at hiring. M3: pay is not set by a schedule. M4: the programme median equals mean log pay among graduates in the employer market, and programmes are large enough that averages of ε_k and of signal noise vanish.

S1 (students). Programmes $j = 1, \dots, J$ with capacities q_j ; applicants with distinct measured abilities $\tilde{a}_k$. All applicants rank programmes in the same strict order $1 \succ 2 \succ \dots \succ J$ and prefer any programme to none; every programme admits by measured ability up to capacity.

A.2 The academic market

Proposition A1 (SpringRank's large-sample limit) *Assume the gravity form, H2, and that the exchange graph (pairs with $\alpha_{uv} + \alpha_{vu} > 0$) is connected. Let $\hat{s}_n$ be the SpringRank scores of A as the ridge vanishes, centred. Then $\hat{s}_n \rightarrow_p \hat{s}$, the centred solution of*

$$\sum_v K_{uv}(\hat{s}_u - \hat{s}_v) = \sum_v K_{uv} \tanh z_{uv}, \quad z_{uv} = \frac{1}{2}(\pi_u - \pi_v) + \frac{1}{2}\{\log \varphi(s_u - s_v) - \log \varphi(s_v - s_u)\},$$

with $K_{uv} = \alpha_{uv} + \alpha_{vu}$. To first order in z , $\hat{s} = c_\varphi s + \frac{1}{2}\pi + \text{const}$ with $c_\varphi = (\log \varphi)'(0)$, so $\hat{s}/c_\varphi = s^* := s + \pi/(2c_\varphi)$ up to a constant. Under SpringRank's own kernel, $\varphi(\Delta) \propto \exp\{-\frac{\beta}{2}(\Delta - 1)^2\}$, $c_\varphi = \beta$ and $z_{uv} = \beta(s_u^* - s_v^*)$ exactly, so the only approximation is $\tanh z \approx z$; where $|z|$ is large (a steep hierarchy or very unequal volumes) $\tanh$ saturates and $\hat{s}$ can misorder some pairs relative to s^* . An exchange between u and v goes from u to v with probability $\{1 + \exp(-2z_{uv})\}^{-1}$, so directions depend on s and π only through z and cannot separate them; under SpringRank's kernel they identify s^* exactly.

Proof. SpringRank minimises $\frac{1}{2} \sum_{uv} A_{uv} (\hat{s}_u - \hat{s}_v - 1)^2$ plus a ridge (De Bacco et al., 2018). Its first-order condition as the ridge vanishes is $\sum_v (A_{uv} + A_{vu})(\hat{s}_u - \hat{s}_v) = \sum_v (A_{uv} - A_{vu})$, a Laplacian system whose centred solution is unique and continuous in A/n on connected graphs. By H2 and the continuous mapping theorem, $\hat{s}_n$ converges in probability to the solution with A/n replaced by α . Under the gravity form ζ cancels from $\alpha_{uv}/\alpha_{vu} = \exp(2z_{uv})$, so $(\alpha_{uv} - \alpha_{vu})/(\alpha_{uv} + \alpha_{vu}) = \tanh z_{uv}$, which gives the display. The odd part of $\log \varphi$ is $c_\varphi \Delta + O(\Delta^3)$ and $\tanh z = z + O(z^3)$, so to first order the right side is $\sum_v K_{uv} \{c_\varphi(s_u - s_v) + \frac{1}{2}(\pi_u - \pi_v)\}$, which $c_\varphi s + \frac{1}{2}\pi$ solves; connectedness makes the solution unique up to a constant. Under SpringRank’s kernel $\log \varphi(\Delta) - \log \varphi(-\Delta) = 2\beta\Delta$ exactly. The direction probability is $\alpha_{uv}/(\alpha_{uv} + \alpha_{vu})$. $\square$

Academic reputation is therefore defined as the revealed position s^* , an estimable parameter of the flow model, so the definition is not circular. Reading it as the academy’s valuation s of the department is what H0 adds: H0 is an identifying normalisation, which only outside data on production and hiring can audit (T0), and those data are proxies for p and h , not the parameters themselves. We do not model the committees’ expectation of doctoral quality.

Lemma A1 (Minimum-violation ordering) *For an ordering ϖ , let $V(\varpi) = \sum_{(u,v): v \succ_{\varpi} u} \mathbb{E}A_{uv}$ be the expected number of hires that move up ϖ , and $V_n(\varpi)$ the sample count. Under H1, every minimiser of V agrees with s on every exchanging pair and on every pair linked by a chain of exchanging pairs that is monotone in s . Under H1 and H2, every minimiser of V_n agrees with s on those pairs with probability tending to one.*

Proof. V is a sum over unordered pairs. For a pair with $s_u > s_v$, an ordering that puts v above u adds $\mathbb{E}A_{uv}$ to V , and the s -ordering ϖ_s adds $\mathbb{E}A_{vu}$. Hence $V(\varpi) - V(\varpi_s) = \sum_{\text{pairs inverted by } \varpi} (\mathbb{E}A_{uv} - \mathbb{E}A_{vu}) \geq 0$, with equality only if ϖ inverts no exchanging pair (H1). A minimiser is a total order, so it also agrees with s along any monotone chain of exchanging pairs; pairs linked by no such chain are left unordered by the data. There are finitely many orderings; by H2, $V_n(\varpi)/n - V(\varpi)/n \rightarrow_p 0$ for each, and $\{V(\varpi) - V(\varpi_s)\}/n$ is bounded away from zero for every ϖ that inverts an exchanging pair, so every sample minimiser (Clauset et al., 2015) agrees with s on those pairs with probability tending to one. $\square$

The lemma is close to a restatement of H1 and concerns the minimum-violation ordering, not the published F ; what links F to s is Proposition A1 together with H0.

Remark A1 (H1, SpringRank and the minimum-violation ordering) (i) If φ is constant (random matching, no valuation), $\mathbb{E}A_{uv} > \mathbb{E}A_{vu}$ if and only if $\pi_u > \pi_v$: H1 holds for the ordering by production relative to hiring. H1 alone therefore does not reveal a valuation; H0 does. (ii) The first-order condition in the proof of Proposition A1 reads $\hat{s}_u = \bar{s}_u + (d_u^{\text{out}} - d_u^{\text{in}})/(d_u^{\text{out}} + d_u^{\text{in}})$, with $\bar{s}_u$ the exchange-weighted mean of u ’s partners: a department sits above its partners by its net-export ratio. (iii) Under H1, F can disagree with the minimum-violation ordering. Take $A_{uv} = 1$, $A_{vu} = 0$, $A_{vw} = 100$, $A_{wv} = 0$, $A_{uw} = 100$ and $A_{wu} = 99$. H1 holds for $u \succ v \succ w$, and the minimum-violation ordering is $u \succ v \succ w$ (99 upward hires; every other ordering has at least 100). SpringRank gives $s_v - s_w = 0.98$ and $s_u - s_w = 0.015$, so $v \succ u \succ w$. On the public edges, F correlates +0.662 on average with a minimum-violation ordering.

Remark A2 (Audits of H0 and H1) T0 replaces F by a degree-corrected Bradley–Terry rank (offsets $\log\{d_u^{\text{out}}d_v^{\text{in}}/(d_v^{\text{out}}d_u^{\text{in}})\}$) and by F residualised on $\log$ doctoral production and the export ratio. Coupling keeps $R_{\text{DC}} = 0.839$ [0.743, 0.936] and $R_{\perp} = 0.598$ [0.471, 0.725] of its size: “partly production”. $1 - R_{\perp}$ bounds the share of coupling carried by the production term of Proposition 1 from above, because residualising also removes the net export that valuation itself produces (H1), but only if F ’s production component lies (linearly) in the span of the two regressors. By Proposition A1 that

component is $\frac{1}{2} \log(p/h)$, which the regressors span only approximately, and observed volumes only proxy p and h ; the audit is to be read with $\log(p/h)$. *T1* asks whether one index orders departments both by where their doctorates are placed and by where their faculty come from: $R = 0.525$ [0.461, 0.583] against a threshold of 0.80, and $R' - R_0 = -0.169$ [-0.229, -0.115] against a benchmark from SpringRank’s own generative model (“not one index”; the benchmark’s calibration was fixed after the literal one had degenerated). The faculty market is two-sided: flows reflect committees’ valuation of degree-granting departments and candidates’ preferences over employers. If committees rank candidates by s_u and candidates rank employers by an index t_v , flows depend on s_u and t_v , and “not one index” reads as $t \neq s$; we do not build that model. A network position or placement power predicts the same agreement of roles, so *T1* cannot separate valuation from position; only *T0* speaks to production.

A.3 The graduate market

Lemma A2 (Attribution weights) *Let the employers of a local market share a record of n_{ig}^e graduates of each programme (i, g) , with $n_i^e = \sum_g n_{ig}^e$ and $w_{ig} = n_{ig}^e/n_i^e$. Measured in field-standardised units, $(\eta_k - \mu_g)/\beta_g$, the name’s pooled record has mean $x_i + \bar{u}_i + \nu_i$, with $\bar{u}_i = \sum_g w_{ig} y_{ig}/\beta_g$ and noise ν_i of variance $\bar{\sigma}^2/n_i^e$, $\bar{\sigma}^2 = \sum_g w_{ig} \sigma^2/\beta_g^2$. Give x_i a normal prior with mean $p_i = \mathbb{E}[x_i | P_i]$ and variance τ_i^2 , and let department deviations, with prior variance τ_y^2 , be learned as deviations from the name’s mean from the $n_{if}^e \delta_f$ graduates attributed to (i, f) (δ_f the attribution probability). Dropping terms in $\bar{u}_i$, which are common to the institution’s programmes and correlate with y_{if} only through its weight w_{if} ,*

$$\hat{x}_i = \omega_i(x_i + \nu_i) + (1 - \omega_i)p_i, \quad \omega_i = \frac{n_i^e \tau_i^2}{n_i^e \tau_i^2 + \bar{\sigma}^2}, \quad \omega_f = \frac{n_{if}^e \delta_f \tau_y^2}{n_{if}^e \delta_f \tau_y^2 + \sigma^2},$$

and the belief about y_{if} is $\omega_f(y_{if} + \nu_{if})$, with ν_{if} the department record’s noise. Hence $\omega_i \geq \omega_f$ whenever $\tau_i^2/\bar{\sigma}^2 \geq \tau_y^2/\sigma^2$ (with $\beta_g \equiv 1$, whenever $\tau_i^2 \geq \tau_y^2$), $\omega_i \rightarrow 1$ as $n_i^e \rightarrow \infty$, and $\omega_f \rightarrow 0$ as $\delta_f \rightarrow 0$.

Proof. A record of n observations of a unit has mean equal to the unit’s value plus noise of variance (per-observation variance)/ n , and normal updating from a prior of variance τ^2 gives the posterior mean $\omega \cdot \text{record} + (1 - \omega) \cdot \text{prior mean}$ with $\omega = n\tau^2/(n\tau^2 + \text{per-observation variance})$. The name’s record pools all n_i^e graduates; a department’s record holds the $n_{if}^e \delta_f$ attributed to it. The map $n\tau^2 \mapsto n\tau^2/(n\tau^2 + c)$ is increasing, and $n_i^e \geq n_{if}^e \delta_f$. $\square$

Institution fixed effects absorb $\bar{u}_i$, and A-orth and the results below hold up to terms of order $\max_g w_{ig}$. The record noise is kept: $\hat{x}$ is then a posterior mean and uncorrelated with its own error $x - \hat{x} = (1 - \omega_i)(x - p) - \omega_i \nu_i$. The noise terms are independent of F , G , S and every latent variable, so they enter no covariance used in Propositions 1, 2 and A2, Corollaries A1–A2 or the ratio of Remark A3; they enter $\text{Var } m(t)$, and hence any coefficient standardised by it.

Lemma A3 (Programme wage) *Under M1–M4, with $R_{if} = \beta_f \hat{x}_i + \omega_f(y_{if} + \nu_{if})$ (Lemma A2),*

$$m_{if}(t) = \mu_f(t) + V_{if}(t) + \beta_f g_i(t), \quad V_{if}(t) = (1 - \lambda_t)R_{if} + \lambda_t \theta_{if}, \quad \lambda_t = b + (1 - b) \frac{t}{t + k},$$

with $b = h_b/(h_0 + h_b)$ and $k = (h_0 + h_b)/h_\varepsilon$. Equivalently $m_{if}(t) - \mu_f(t) = \beta_f M_i(t) + \Lambda_{ft} y_{if} + (1 - \lambda_t) \omega_f \nu_{if}$, with $M_i(t) = (1 - \lambda_t) \hat{x}_i + \lambda_t x_i + g_i(t)$ and $\Lambda_{ft} = \omega_f + (1 - \omega_f) \lambda_t$; the university-wide part of pay is $m_{if}^U(t) = \beta_f M_i(t)$.

Proof. After the hiring signal and t yearly signals, normal updating gives the posterior mean $(1 - \lambda_t)R_{if} + \lambda_t \bar{z}_k$, where $\bar{z}_k$ is the precision-weighted mean of k 's signals, which is η_k plus noise, and $\lambda_t = (h_b + th_\varepsilon)/(h_0 + h_b + th_\varepsilon)$; direct algebra gives $\lambda_t = b + (1 - b)t/(t + k)$. By M1(iii) the wage adds the known growth. Averaging over the programme's graduates removes ε_k and the signal noise (M4), and M4 equates the median with the mean; the average posterior mean is $V_{if}(t)$. Substituting $\theta_{if} = \beta_f x_i + y_{if}$ and R_{if} and collecting the coefficient of y_{if} , $(1 - \lambda_t)\omega_f + \lambda_t = \Lambda_{ft}$, gives the second form. $\square$

Employer reputation. $V_{if}(t)$, the graduate market's mean belief about the productivity of the programme's graduates at t , equals $m_{if}(t) - \mu_f(t) - \beta_f g_i(t)$ under M1–M4, so it is revealed at every t once growth is known. *Entry reputation* is $R_{if} = V_{if}(0)$; since $g_i(0) = 0$, at entry with $b = 0$ pay reveals it. The *reputation premium* $R - \theta$, the object of MacLeod et al. (2017), decays with learning, $V(t) - \theta = (1 - \lambda_t)(R - \theta)$. At a name with $\omega_i = 1$, $V - \theta = -(1 - \lambda_t)(1 - \omega_f)y_{if}$ up to record noise, so the university-wide parts of V and θ coincide and pay at the name ranks selection-inclusive expected productivity. In the data, employer reputation is programme earnings as priced by the graduate market, selection included; it differs from mean productivity only where $\omega < 1$ or at entry, and these data measure neither: M4 fails most at one year (Remark A3). Under M1 with a public and costless record, every graduate ever observed would enter the record, and ω_f would be close to one unless τ_y^2 were tiny, which is itself a statement of little departmental content; a small ω_f needs costly attribution or segmented records.

College reputation in economics. MacLeod et al. (2017) measure a college's reputation by the mean admission score of its graduates, find that it is correlated with graduates' earnings growth, and find that a new public signal of individual skill, a college exit exam, lowered its return. The within-band rise found here (T4: the partial correlation of pay with selectivity given G rises by $+0.0266$ [$+0.0157$, $+0.0374$] per year within fixed cohorts, on average over bands, concentrated in the middle bands) is in line with the first finding. *Growth.* If growth is complementary with ability, $g_i(t) = \gamma_t x_i$ with $\gamma_0 = 0$, it loads on the same x that sets the level, selectivity included, which is in line with T4; this is one specification of the growth slot, not a test of it, and M1(iii) then requires x_i to be known, as it is where $\omega_i = 1$.

A.4 Students

Choice. Student k , with ability a_k and measured ability $\tilde{a}_k$, values programme p at $u_k(p) = \mathbb{E}_k[w_p(a_k) \mid \Sigma_k] + \varepsilon_{kp}$, where Σ_k are the signals k sees (rankings of the academic-reputation kind, published programme earnings, intake statistics) and ε_{kp} is taste. Write U_p for the part common to all students, the programme's reputation among students. Programme p admits applicants with $\tilde{a}_k \geq c_p$, and cutoffs clear capacity, so c_p and intake ability ξ_p are equilibrium outcomes. S1 is the case $\varepsilon \equiv 0$: then students rank programmes by U_p , and by Lemma A4 ξ_p is an increasing function of the rank of U_p . With tastes, comparative advantage or strategic application the ordering holds only on average, and self-selection on gains (Kirkebøen et al., 2016) breaks it. Which of academic reputation, employer reputation, peer ability and value added U_p loads on is the next study's question; for schools, Abdulkadiroğlu et al. (2020) ask whether parents' choices follow school effectiveness or peer quality.

Lemma A4 (Selectivity as a revealed ranking) *Under S1 the stable matching is unique (Gale and Shapley, 1962). It assigns the q_1 applicants of highest measured ability to programme 1, the next q_2 to programme 2, and so on. Every entrant of programme j then has higher measured ability than every entrant of programme $j + 1$, so the mean, the median and every other location statistic of measured intake ability order programmes as students' common ranking does. If the distribution*

of a given $\tilde{a}$ is increasing in $\tilde{a}$ in the sense of first-order stochastic dominance (implied by the monotone likelihood ratio property, and by jointly Gaussian a and $\tilde{a}$ with positive correlation), and admission depends on a only through $\tilde{a}$, the mean and every quantile of true intake ability order programmes in the same way.

Proof. Existence: in the serial assignment μ^* , an applicant k who prefers j to $\mu^*(k)$ has j ranked above $\mu^*(k)$; j was filled before k was placed, and only with applicants of higher ability than k , so j does not want k , and μ^* has no blocking pair. Uniqueness, by induction on J : let μ be stable and let k be among the q_1 highest-ability applicants. If $\mu(k) \neq 1$, then k prefers programme 1, and programme 1 either has a free seat or holds an applicant of lower ability than k ; either way $(k, 1)$ blocks μ . So programme 1 holds exactly the q_1 highest-ability applicants. Removing them and programme 1 leaves a stable matching of the smaller market, to which the argument applies again. The ordering claim follows because the entrants of j are all placed before those of $j + 1$. For true ability, the distribution of a among entrants of j averages $\Pr(a \leq z \mid \tilde{a})$ over measured abilities that all exceed those of $j + 1$; each such term is no larger, so the distribution for j dominates that for $j + 1$ at every z . $\square$

S1 is a sufficient condition, and it is strong. Admission and yield rates can be manipulated, which is why Avery et al. (2013) rank colleges from the matriculation choices of students admitted to several. A university $\times$ field ordering by students exists only where admission is by programme (UK UCAS by course, Norway, Chile, China’s national examination with cutoffs by institution and major). US students are admitted to an institution and choose a major later, sometimes under major-specific grade restrictions (Bleemer and Mehta, 2022), so the US indicators (mean SAT, admission rate) are university-level: a capacity-weighted average over the institution’s programmes, with any field-specific part of students’ ranking in y_{if} . UK LEO earnings by provider $\times$ subject $\times$ prior-attainment band carry a programme-level measure of intake.

A.5 Proof of Proposition 1

Reliability. By E, $\text{Cov}(F, Y) = \text{Cov}(s^*, m)$, $\text{Var } F = \text{Var } s^* / \text{rel}_F$ and $\text{Var } Y = \text{Var } m / \text{rel}_Y$, so on normal scores $\text{Corr}(F, Y) = \sqrt{\text{rel}_F \text{rel}_Y} \text{Corr}(s^*, m)$; by A0, ρ_f is a strictly increasing function of it.

Decomposition. By Lemma A3, $m - \mu_f = m^U + \Lambda_{ft}y$ plus record noise. Write $s^* = \alpha_f a + r + \pi / (2\beta)$ and project on ξ : $s^* = \{\text{Cov}(s^*, \xi) / \sigma_\xi^2\} \xi + s^{*\perp\xi}$ with $s^{*\perp\xi} = \alpha_f a^{\perp\xi} + r + \pi^{\perp\xi} / (2\beta)$, since $\text{Cov}(r, \xi) = 0$ (A-orth). By A-orth, $\text{Cov}(a, y) = \text{Cov}(\xi, y) = 0$, so $\text{Cov}(a^{\perp\xi}, m) = \text{Cov}(a^{\perp\xi}, m^U)$ and $\text{Cov}(\xi, m) = \text{Cov}(\xi, m^U)$; and $\text{Cov}(r, m^U) = 0$, so $\text{Cov}(r, m) = \Lambda_{ft} \text{Cov}(r, y) = \Lambda_{ft} \kappa_f \sigma_r \sigma_y$. Summing gives (3).

Partialling selectivity. The partial covariance of F and Y given S is

$$\text{Cov}(F, Y) - \text{Cov}(F, S) \text{Cov}(S, Y) / \text{Var } S.$$

By E, $\text{Cov}(F, S) = \text{Cov}(s^*, \xi)$, $\text{Cov}(S, Y) = \text{Cov}(\xi, m)$ and $\text{Var } S = \sigma_\xi^2 / \text{rel}_S$, so the subtracted quantity is rel_S times the first term of (3). The partial therefore exceeds the last three terms, that is, bounds agreement beyond measured intake from above, if and only if $\text{Cov}(s^*, \xi)$ and $\text{Cov}(\xi, m)$ have the same sign. $\square$

Four remarks on use. (i) $\text{Cov}(s^*, m) > 0$ is not implied: it needs the two valuations to load in the same direction on the university-wide components, and κ_f and the production term not so negative as to offset. (ii) The reported figures (+0.43 to +0.14) are mean partial correlations. A partial correlation divides the partial covariance by residual standard deviations below the raw ones, so the share of covariance that runs through ξ is larger than the proportional fall in correlation. (iii) e_S contains $w_{if} y_{if}$, so the partial is exact only up to terms of that order (E). (iv) The first

term is not selection alone: peer quality and resources that rise with intake load on ξ , which is why Dale and Krueger (2002) treat selectivity as a candidate treatment; programme aggregates cannot separate a graduate's own ability from her peers', because at the programme mean they coincide. If students sort on reputation (MacLeod and Urquiola, 2015), intake is partly an outcome of s^* and R , and partialling it removes the part of agreement that runs through attracting students. The UK attainment-band comparisons control graduates' own prior attainment, in the sense of MacLeod et al. (2017); the US partial on institution-mean SAT removes peer and resource effects as well, so the two are different objects.

Corollary A1 (Fields as scales of one index) *Suppose that (a) λ_t is common across fields, so that $m_{if}^U(t) = \beta_f M_i(t)$ with $M_i(t)$ the same in every field (Lemma A3); (b) α_f , σ_r and the joint moments of (a, ξ, M) over $\mathcal{I}_f$ are common to all fields; (c) $\kappa_f \approx 0$ in covariance, while the field-specific variance $\Lambda_{ft}^2 \sigma_{y,f}^2$, the record noise and e_Y may differ across fields; and (d) H0. Then the true-score coupling of field f and the true-score correlation of its pay with ξ are both proportional to*

$$\frac{\beta_f}{\text{sd } m_f} = \left\{ \text{Var } M + \frac{\Lambda_{ft}^2 \sigma_{y,f}^2 + \text{noise variance}}{\beta_f^2} \right\}^{-1/2},$$

with a ratio, $\alpha_f \text{Cov}(a, M) \sigma_\xi / \{\text{Cov}(\xi, M) \text{sd } s\}$, that is the same in every field. The two field maps are therefore collinear, and both order fields by the share of programme pay variance that is university-wide, not by the steepness of pricing; the same holds with any status index of this form in place of F .

Proof. By (c), $\text{Cov}(s, m) = \alpha_f \beta_f \text{Cov}(a, M)$, and $\text{Cov}(\xi, m) = \beta_f \text{Cov}(\xi, M)$ by A-orth. Dividing by $\text{sd } s \cdot \text{sd } m_f$ and $\sigma_\xi \text{sd } m_f$ gives two correlations proportional to $\beta_f / \text{sd } m_f$, and $\text{sd}^2 m_f = \beta_f^2 \text{Var } M + \Lambda_{ft}^2 \sigma_{y,f}^2 + \text{noise variance}$. By (b), the ratio and $\text{sd } s = (\alpha_f^2 + \sigma_r^2)^{1/2}$ do not depend on f . $\square$

The assumption that fields differ only in the scale at which they price a common university-level index comes close to the conclusion. The observed true-score correlation of the two maps, +0.94, is a consistency check, not a prediction; replacing F by three non-academic indices (T5) is inconclusive.

A.6 Proof of Proposition 2

Definitions and identity. Standardise the ranks of F , G and Y . Regress F 's standardised ranks on G 's: the slope is $\cos \vartheta = \rho(F, G)$ and the residual is $\sin \vartheta H$, with H of unit variance and orthogonal to G . sr_F is the Pearson correlation of Y 's standardised ranks with H , not a Spearman correlation of the residual, whose re-ranking would break the identity. By linearity, $c_F = c_G \cos \vartheta + \text{sr}_F \sin \vartheta$, and in the regression of Y on F and G the coefficient on F is $\text{sr}_F / \sin \vartheta$ (Frisch–Waugh). Then $c_F - c_G = \text{sr}_F \sin \vartheta - c_G(1 - \cos \vartheta)$, which is positive if and only if $\text{sr}_F > c_G(1 - \cos \vartheta) / \sin \vartheta = c_G \tan(\vartheta/2)$ for $\vartheta \in (0, \pi)$. The identity holds for any three variables; the model enters only in reading sr_F and c_G .

Model reading. “ $G = a$ up to negligible error” requires, besides $\text{rel}_G \approx 1$, that the published G , which pools the hires of all fields including f , carry a negligible hire-weighted mean of field-specific deviations. That holds if field deviations are independent within an institution; if they are correlated (a school layer, as in the SI), $G = a + \bar{r}^S$ with $\bar{r}^S$ not negligible and $F^\perp G \neq r + e_F$. On normal scores with $G = a$ ($\text{Var } a = 1$) and H0, $F = \alpha_f a + r + e_F$, so $\cos \vartheta = \alpha_f / \text{sd } F$ and $F^\perp G = r + e_F$, whose variance is $\sin^2 \vartheta \text{Var } F$. All of F 's error lies in $F^\perp G$, since e_F is independent of G ; hence the reliability of $F^\perp G$ is $\text{rel}_H = \sigma_r^2 / (\sigma_r^2 + \sigma_{e_F}^2) = 1 - (1 - \text{rel}_F) / \sin^2 \vartheta$, below rel_F whenever $\cos \vartheta \neq 0$ and $\text{rel}_F < 1$. By Lemma A3 and A-orth, $\text{Cov}(Y, F^\perp G) = \text{Cov}(m, r) = \Lambda_{ft} \kappa_f \sigma_r \sigma_y$,

so $\text{sr}_F = \sqrt{\text{rel}_H} \Lambda_{ft} \kappa_f \sigma_y / \text{sd } Y$, and $c_G = \text{Cov}(a, m^U) / \text{sd } Y$ because $\text{Cov}(a, y) = 0$. Agreement concentrates at the university level ($c_G \geq c_F$) if and only if $\sqrt{\text{rel}_H} \Lambda_{ft} \kappa_f \sigma_y \leq \tan(\vartheta/2) \text{Cov}(a, m^U)$. Without H_0 , $F^{\perp G}$ also carries $\pi^{\perp a} / (2\beta)$. $\square$

At the across-field means ($\cos \vartheta = 0.786$, $c_G = 0.450$) the threshold is 0.156 against $\text{sr}_F = 0.077$, exceeded in 15 of 57 fields; on true scores the thresholds are 0.13 and 0.17 (lower-bound and extrapolated reliability of F) against about 0.12 and 0.09. The within-institution slope, $+0.0084$ $[+0.0000, +0.0164]$ log points per SD of F $[-0.0012, +0.0180]$ under the two-way variance; about zero with one- and five-year medians), is individually clear only in computer science.

Entry and attribution. By Lemma A3, at $t = 0$ with $b = 0$ the field-specific term is $\omega_f \kappa_f \sigma_r \sigma_y$, non-zero if and only if $\kappa_f \neq 0$ and $\omega_f > 0$. For $t > 0$ or $b > 0$, $\Lambda_{ft} \geq \lambda_t > 0$, and the term is non-zero if and only if $\kappa_f \neq 0$. With learning speeds estimated elsewhere (Lange, 2007; Arcidiacono et al., 2010) ($b = 0$, $k = 3$ gives $\lambda_4 = 0.57$), a small four-year term would mean little content; small attribution explains it only if learning is much slower ($b \approx 0$, $k \gtrsim 20$). Nothing in these data measures learning speed, so the two readings are not separated, and no mechanism is claimed.

A.7 Years since graduation and pay schedules

Proposition A2 (Years since graduation and pay schedules) (a) For any indicator Z ,

$$\text{Cov}\{Z, m(t)\} = \text{Cov}(Z, R) + \lambda_t \text{Cov}(Z, \theta - R) + \beta_f \text{Cov}\{Z, g(t)\}.$$

The middle covariance is zero if Z is in employers' information at hiring and R is the rational expectation of θ given that information, or if Z is university-wide and $\omega_i = 1$. In the second case learning moves $\text{Cov}\{F, m(t)\}$ only by $(1 - \omega_f)(\lambda_t - \lambda_0) \kappa_f \sigma_r \sigma_y$. (b) Let a share ψ_p of programme p 's graduates be paid on a schedule at level ℓ_p , and write $\psi_p = \bar{\psi}_f + \tilde{\psi}_p$, $\ell_p = \bar{\ell} + \tilde{\ell}_p$ and $D = \bar{m}^{\text{mkt}} - \bar{\ell}$. To first order, on normal scores,

$$\text{Corr}(F, Y) = \sqrt{\text{rel}_F \text{rel}_Y} \frac{(1 - \bar{\psi}_f) \text{Cov}(s^*, m^{\text{mkt}}) + \bar{\psi}_f \text{Cov}(s^*, \ell) - D \text{Cov}(s^*, \psi)}{\text{sd } s^* \cdot \text{sd } m},$$

with $\text{Var } m = (1 - \bar{\psi}_f)^2 \text{Var } m^{\text{mkt}} + \bar{\psi}_f^2 \text{Var } \bar{\ell} + D^2 \text{Var } \tilde{\psi}$ plus cross terms and $\text{rel}_Y = \text{Var } m / (\text{Var } m + \text{Var } e_Y)$. If ℓ and ψ do not vary across programmes, the true-score coupling equals the market's, $\text{Corr}(s^*, m^{\text{mkt}})$.

Proof. (a) By Lemma A3, $m(t) - \mu_f(t) = R + \lambda_t(\theta - R) + \beta_f g(t)$, which gives the covariance decomposition by linearity. If $R = \mathbb{E}[\theta \mid \mathcal{J}]$ for employers' information $\mathcal{J}$ at hiring and Z is $\mathcal{J}$ -measurable, then $\mathbb{E}[Z(\theta - R)] = \mathbb{E}\{Z \mathbb{E}[\theta - R \mid \mathcal{J}]\} = 0$ and $\mathbb{E}(\theta - R) = 0$, so $\text{Cov}(Z, \theta - R) = 0$. This R is not the R of Lemma A2 in general: if F is public and $\kappa_f \neq 0$, the rational R contains $\mathbb{E}[y \mid F]$, while Lemma A2 credits only $\omega_f y$. If instead $\omega_i = 1$, then $\hat{x} = x$ up to record noise and $\theta - R = (1 - \omega_f)y$, which is uncorrelated with every university-wide Z by A-orth. For $Z = F$, $\text{Cov}(F, \theta - R) = (1 - \omega_f) \kappa_f \sigma_r \sigma_y$, and learning contributes $(\lambda_t - \lambda_0)(1 - \omega_f) \kappa_f \sigma_r \sigma_y$. (b) Treat the median as mean log pay (M4; the median of a mixture is not linear in ψ). Then $m_p = (1 - \psi_p) m_p^{\text{mkt}} + \psi_p \ell_p$; expanding and dropping products of deviations, $m_p = \text{const} + (1 - \bar{\psi}_f) \tilde{m}_p^{\text{mkt}} + \bar{\psi}_f \tilde{\ell}_p - D \tilde{\psi}_p$. Taking the covariance with s^* and the variance, and applying E, gives the display. With $\bar{\ell} = \bar{\psi} = 0$, m is $(1 - \bar{\psi}_f) m^{\text{mkt}}$ plus a constant, and correlations are scale-free. $\square$

Learning and the career rise. A name's record pools the graduates of all its programmes (Lemma A2), so under rational expectations ω_i is near one wherever employers see an institution's graduates often; there, learning about individuals moves no university-wide loading, and the absence of a decline in the status loading net of selectivity (0 of 9 tests), the signature sought after

Altonji and Pierret (2001), is close to guaranteed and weak evidence about learning. Within fixed cohorts, coupling rises by +0.031 per year [+0.021, +0.041] in the US and +0.026 in the UK. At names with $\omega_i \approx 1$, a rise of the university-level covariance needs growth; learning can add through the department term, bounded by $(1 - \omega_f)(\lambda_t - \lambda_0)\kappa_f\sigma_r\sigma_y$ and small given sr_F , and through rarely seen names ($\omega_i < 1$); on the correlation scale, a fall in pay variance unrelated to F also raises coupling. T3, a null, does not separate growth from learning about rarely seen names.

Pay schedules. A schedule that only rescales market pay leaves every correlation unchanged, including the correlation of pay with ξ (the scaling of β_f by $1 - \bar{\psi}_f$ cancels). It lowers observed coupling only through rely_r , since programme signal variance shrinks by $(1 - \bar{\psi}_f)^2$ while sampling noise need not, and through variance of $\tilde{\ell}$ and $\tilde{\psi}$ unrelated to F (regional allowances, sorting into scheduled jobs unrelated to status); lower market agreement lowers it too. Under a national schedule $\tilde{\ell} = 0$ at entry; later $\ell_p(t) = \sum_g \Pr(\text{grade}_t = g \mid p) \ell_g$, which covaries with F only if grade attainment does, and exit from scheduled jobs that depends on status moves $\tilde{\psi}$ in the same way. The prediction can be checked by disattenuating the coupling of scheduled fields with their median reliabilities. In the US Scorecard (`scripts/73`), special education couples at +0.55 with reliability 0.887, and teacher education (subjects), with reliability 0.93, is classed decoupled (coupling below 0.25), so their contrast is not a reliability difference; within (b) the weak coupling needs schedule variance unrelated to F or lower market agreement. The UK subjects paid on national scales (nursing, medicine and teaching, +0.120 against +0.523) have no reliabilities here. Where M3 fails the wage stops revealing the market's valuation. If employers ration jobs and promotions, their valuation acts on quantities; graduates' own choices produce the same quantities, so reading them as valuation needs rationing. T6 supports the quantity prediction (+0.238 [+0.065, +0.380]), though its index largely records location. Grade progression and exit were added after the rise of UK teaching coupling had been seen.

Corollary A2 (Loadings net of selectivity) *Let $e_i = x_i - p_i$ be the error of employers' public prior about the name. In a linear projection of $m^U(t)$ on (ξ, G) , the change of the coefficients that is due to learning about individuals is $\beta_f \lambda'_t c$, where c are the coefficients of the projection of $(1 - \omega_i)e_i$ on (ξ, G) ; with ω_i common across institutions, $c = (1 - \omega)c_e$ with c_e the projection coefficients of e . The status loading net of selectivity therefore moves through learning only if $\omega_i < 1$ for some names and $c_G \neq 0$, and with $\beta_f > 0$ it declines if $c_G < 0$. That is the case, for example, when G is part of employers' public signal and intake ability is not, and both e and G covary positively with ξ (then $\text{Cov}(e, G) = 0$ and $c_G \propto -\text{Cov}(e, \xi) \text{Cov}(\xi, G)$), the setting of Altonji and Pierret (2001).*

Proof. $x - \hat{x} = (1 - \omega_i)e - \omega_i\nu$, so $m^U = \beta_f\{x - (1 - \lambda_t)[(1 - \omega_i)e - \omega_i\nu] + g\}$. Projection is linear, ν projects to zero on (ξ, G) , and only the term in e depends on λ_t through learning; differentiate in t . $\square$

Remark A3 (The entry pattern) *On the PSEO fixed cohorts, in standardised rank regressions of pay on F and G within field-by-cohort cells, $b_F = +0.189 [+0.060, +0.309]$, +0.176 and +0.177 at one, five and ten years, and $b_G = -0.013 [-0.121, +0.092]$, +0.166 and +0.328. Standardisation divides both by $\text{sd } Y_t$, which learning itself changes, so single coefficients cannot be compared across horizons. The name loading $N = \text{Cov}(G, Y_t)/\text{sd } Y_t = b_G + \cos \vartheta b_F$ and b_F share that scale, so N/b_F is free of it. Suppose learning is the only source of change (no growth, migration or change in who is observed), $H0$ holds, $G = a$, $\beta_f > 0$, $\kappa_f > 0$, $\omega_i \geq \omega_f$ (Lemma A2), $C = \text{Cov}(a, x) \geq 0$ and $\text{Cov}(a, p) \geq 0$ (the public prior does not load negatively on status). By Lemma A3 and A-orth, before standardisation $N \propto \beta_f\{(1 - \lambda_t)C_{\hat{x}} + \lambda_t C\}$ with $C_{\hat{x}} = \text{Cov}(a, \hat{x}) = \omega_i C + (1 - \omega_i)\text{Cov}(a, p)$,*

and $b_F \propto \Lambda_{ft} \kappa_f \sigma_r \sigma_y / \text{Var}(F^{\perp G})$, with factors that do not depend on t . Differentiating,

$$\frac{d}{d\lambda_t} \frac{(1 - \lambda_t)C_{\hat{x}} + \lambda_t C}{\omega_f + (1 - \omega_f)\lambda_t} = \frac{\omega_f C - C_{\hat{x}}}{\Lambda_{ft}^2} \leq 0,$$

since $C_{\hat{x}} \geq \omega_i C \geq \omega_f C$. Under learning alone N/b_F therefore cannot rise with t (the umbrella ordering of the SI). It does: $N/b_F = \cos \vartheta + b_G/b_F$, and b_G/b_F goes from negative at one year to above one at ten, whatever $\cos \vartheta$. Four readings remain, and the data do not choose among them: employers know departments from entry and the name loading rises through growth; λ_t is nearly constant over these years, because ability is revealed at hiring (Arcidiacono et al., 2010) or learning is negligible; F tracks a trait other than the field-specific term, such as programme-level selection; or the first year is an artefact, since graduate study and coverage that falls with prestige depress the first-year medians of high-status institutions (M_4 fails most at one year), pushing b_G towards zero. Because M_4 fails there, the reading of entry pay as entry reputation cannot be checked on these data.

A.8 At an admission cutoff

Corollary A3 (At an admission cutoff) Let graduate k of programme $p = (i, f)$ have productivity $\eta_k = \beta_f a_k + \delta_p + u_k$, with a_k her ability, δ_p the programme's effect on productivity (value added, including peer and resource effects) and u_k idiosyncratic, so that $\theta_p = \beta_f \bar{a}_p + \delta_p$ with $\bar{a}_p$ the mean ability of p 's graduates. Let pay follow M1 with symmetric learning, $w_k(t) = \mu_f(t) + (1 - \lambda_t)R_p + \lambda_t \eta_k + \beta_f g_p(t)$ plus noise, with R_p employers' belief about θ_p at hiring and $g_p(0) = 0$ growth attributable to the programme. At an admission cutoff c_p in measured ability, with a_k and u_k continuous at c_p , the effect of admission to p for a student whose fallback is p' in the same field is

$$\tau(t) = \Delta\delta + (1 - \lambda_t)(\Delta R - \Delta\delta) + \beta_f \Delta g(t),$$

with $\Delta X = X_p - X_{p'}$. If employers know both programmes' means ($R = \theta$), the second term is $(1 - \lambda_t)\beta_f \Delta \bar{a}$. If p' is in another field f' , add $\Delta\mu(t)$ and $\lambda_t(\beta_f - \beta_{f'})a_k$, a comparative-advantage term.

Proof. Compare the expected pay of the same student in p and in p' ; continuity at the cutoff makes the students just above and below comparable. a_k and u_k cancel within a field, leaving $(1 - \lambda_t)\Delta R + \lambda_t \Delta\delta + \beta_f \Delta g(t)$, which rearranges to the display. With $R = \theta$, $\Delta R - \Delta\delta = \beta_f \Delta \bar{a}$. Across fields the field effects and the prices of a_k differ. $\square$

Four consequences. (i) The return at a cutoff contains a reputation rent, $(1 - \lambda_t)(\Delta R - \Delta\delta)$, which employers pay even if it is pure signal (at known names, the gap in mean ability of the student's classmates) and which learning erodes. (ii) As $\lambda_t \rightarrow 1$, $\tau(t) \rightarrow \Delta\delta + \beta_f \Delta g(t)$: in the long run the estimand is value added plus growth. (iii) The profile of $\tau(t)$ over years since graduation is an individual-level test of employer learning in the sense of Altonji and Pierret (2001), which programme aggregates cannot supply (Proposition A2a); a decline indicates a rent, while a flat profile does not rule one out, since $\Delta g(t)$ can rise. (iv) Even with $\Delta\delta = 0$, admission to a more selective programme raises pay over the horizons where $\lambda_t < 1$, so a student who values selectivity is not thereby irrational; whether students' choices load on $\Delta \bar{a}$, on $\Delta\delta$ or on the reputations is testable. Fallbacks differ across students, so the comparison is between university $\times$ field programmes (Kirkeboen et al., 2016); Hastings et al. (2013) estimate returns at degree-programme admission cutoffs, the object closest to ours, and Mountjoy and Hickman (2021) separate relative value added from match effects. In Anelli (2020) part of the enrolment effect runs through changed

field choice, which in the corollary is the cross-field terms, and the channels he reports (degree completion, on-time graduation, peer quality) sit in δ and $\bar{a}$.

A.9 A design for the next study

(1) Estimate $\tau_p(t)$ at many programme cutoffs, in a system that admits by programme and links admission to earnings by years since graduation. (2) Describe each marginal student's fallback p' from her next-best choice (Kirkebøen et al., 2016). (3) Project $\tau_p(t)$, horizon by horizon, on the target-minus-fallback gaps in academic reputation (ΔF), employer reputation (ΔY) and peer ability ($\Delta \bar{a}$ or $\Delta \xi$). Under Corollary A3 the loading on peer ability falls with t if the rent is real, and academic reputation enters only through its correlation with $\Delta \delta$, ΔR or Δg , since the model has no channel from s to pay. (4) Relate students' application orderings and cutoffs to the same gaps, which asks which reputation students follow. Three questions for researchers with cutoff data: (a) in data like Anelli (2020), does the cutoff effect change with years since graduation in a way that separates a reputation rent from value added? (b) Can the distribution of fallbacks at a cutoff be described by the fallback programmes' two reputations? (c) Which public or accessible data combine programme-level cutoffs with earnings by years since graduation?

A.10 Predicted, accommodated and in tension

No result follows for every admissible parameter value. One follows under a named added assumption: if fields differ only in the scale at which they price a common university-level index, the field maps of coupling and selectivity pricing are collinear (Corollary A1; true-score $r = +0.94$, a consistency check). Accommodated by parameter values inferred from the same results: agreement at the name and with intake; F not beating G ; the absent decline; weak coupling under national pay schedules, through lower reliability or schedule variance unrelated to F (not through scaling); and strong coupling despite schedules in US special education (+0.55, reliability 0.887) and UK allied health (+0.570), which (b) allows where reliability is high and schedule variance unrelated to F is small, and which is therefore no longer listed as a tension (UK reliabilities are not measured here). Accommodated with an added term: the career rise, by growth, learning about rarely seen names or falling variance unrelated to F . In tension, under the conditions of Remark A3: the entry pattern.

A.11 Scope

The model predicts no magnitudes. Its parameters β_f , κ_f , ω_f , ω_i , growth, ψ_f and the loadings of the university-wide components are free, and several are inferred from the results they organise. At the programme level, selection and value added are bundled in ξ , v and y : $\beta_f v_i$ holds unmeasured ability, priced at β_f , together with value added, which need not be priced at β_f . Nothing here says what attending a programme does to pay; Corollary A3 separates the two, in productivity units, only at a cutoff. Growth is a reduced-form slot for networks, skills, promotion ladders and sorting. Reputations are static, so calendar and cohort drift and entry conditions lie outside the model. All comparisons are across institutions within a field, so the model is silent on the choice of field. It uses one index per market and a linear-Gaussian form, so it cannot represent premia concentrated at the top of the distribution, which medians miss. The UK hierarchy has no department axis, so Proposition 2 applies to the US only.